# CheckIn

## Daily Work Report — VLE Redesign & System Flow

Student Attendance System · IDE201 Capstone · Group 2 · College of Science and Technology

| | |
|---|---|
| Contributor | Ugyen Norbu (GitHub: UgayNobu) |
| Role | Database manager; backend; repository owner |
| Date | Friday, 23 October 2026 |
| Repository | github.com/UgayNobu/Checkin (branch: feature/vle-theme) |
| Status | Finished and committed on my laptop. Not pushed yet: waiting for the team's approval, then a pull request into dev. |

## 1. Summary

Today I redesigned the four dashboards (student, tutor, admin and management) to look like the college VLE (Moodle), without changing what any page does and without touching the landing, login and sign-up pages. After that I worked through the team's feedback page by page: tutor email notifications now actually send, My Modules, Attendance Records, Reports & Analytics and the absence-request page were rebuilt, two Semesters bugs were fixed, and light colour accents were added so pages are no longer plain white.

I also made attendance fully timetable based: a class that the tutor never starts is now cancelled automatically, and classes of modules that have no tutor yet no longer block rooms. Section 3 describes the full system flow (attendance, and tutor–admin interaction) as the team asked for the sprint.

| 10 | 4 | 52 | 0 | 2 |
|---|---|---|---|---|
| commits | dashboards redesigned | files changed | database changes | system flows documented |

## 2. Work completed

### 2.1 VLE look for all four dashboards

The logged-in app now follows the Moodle "Boost" style used by the CST VLE. All the CSS is scoped to the logged-in shell, so the landing, login, sign-up and forgot-password pages keep their own design.

- **Top navigation bar:** logo, the role's main links with a blue underline on the active page, a "More" menu for the rest, and a user menu (Profile, Log out) on the right.
- **Phones:** below 1100 px the links move into a slide-in drawer opened by a menu button.
- **Pages:** every page gets a large title with a one-line description and a footer. Cards, buttons, inputs and striped tables use the Moodle colours (primary blue #0F6CBF, text #1D2125, borders #DEE2E6).
- **Same features:** only the look changed; no page lost or gained a feature (apart from the items asked for below).

## 2.2 Tutor email notifications now work

The tutor Settings page showed Email, SMS and Push switches, but nothing was ever sent. Now there is one “Email notifications” switch, and emails go to the address the tutor signed up with. SMS and Push were removed because the system has no SMS gateway or push service, so the switches did nothing.

| Event | Who gets the email |
|---|---|
| Password changed or reset | The account owner |
| Absence request approved or rejected by management | The student and every tutor of the student’s modules |
| Attendance below 90% (checked once a day) | The student, and one email per module to the tutor listing the students; nobody is emailed again within 7 days |
| Tutor account approved by the admin | The new tutor |

Turning the switch off stops the emails; in-app notifications still appear. The daily low-attendance check now runs automatically from the server.

## 2.3 My Modules

- **Course cards:** each module is a VLE-style card with a coloured pattern image, code, name and number of students.
- **Module page:** a “Self enrolment” box (enrolment key, Allow self enrolment tick box in its own row, Save changes) and an “Enrolled students” table with Enrol student and Remove. This fixed the tick box sitting under the key and the misaligned key field.

## 2.4 Attendance Records

Because a tutor teaches at most three modules, the list is now one wide card per module showing: students, classes held and hours, last class, average attendance with a bar, and how many students are below 90%. “View attendance” opens the module.

## 2.5 Reports & Analytics

- **Headline numbers:** overall attendance, classes held, students below 90% and flagged check-ins.
- **Weekly attendance chart:** one line per module for the last 6 weeks with a dashed 90% line. Weeks without a class are left as gaps, not shown as 0%. Hovering a week shows each module’s % and number of classes.
- **Modules:** each module’s semester % against the 90% mark; clicking opens its records.
- **Students who need attention:** everyone below 90%, lowest first; clicking a row opens that student’s record. The page refreshes every 30 seconds.

## 2.6 Absence request (medical leave) page

The detail page now has two cards: on the left the student, a status badge (Pending, Approved or Rejected), module, date missed, when it was submitted and who reviewed it, plus the reason; on the right a large preview of the document with an “Open in new tab” button. Students, tutors and management all use this page.

## 2.7 Semesters and the user menu

- **Bug — “Past” semester:** Spring 2027 showed as Past because the page called any semester Past once its start date had gone by. Now only an ended semester is Past; one whose dates have begun but that hasn’t been started shows “Not started”.
- **Overlapping dates blocked:** the server now refuses a new semester whose dates overlap an existing one (Spring 2027 had been entered as 19 Oct 2026 – 19 Jan 2027, overlapping Autumn 2026). The wrong row has to be corrected once in Supabase.
- **Profile link:** admin and management have no profile page, so the Profile item in their user menu was removed (it only went back to the dashboard).

## 2.8 Timetable-based classes

- **Automatic cancellation:** if a tutor does not start attendance before a class ends, the class becomes “Cancelled – Not started by the tutor”. Students see it as cancelled and are not marked absent; management sees “Automatic” in Class changes. The server checks every 5 minutes and whenever the timetable is opened, looking back 7 days. Only modules with a tutor are affected.
- **Modules not assigned yet:** their classes are shown greyed out as “Module not assigned” with “Room … (free to book)”. They no longer block the room, the tutor or the students, so extra classes can be booked and attendance taken in those slots during testing.

## 2.9 Colour accents, programme cards and the Teachers table

- **Colour on every dashboard:** a blue strip on the navigation bar, a soft blue band behind page titles, a dark-blue footer, light-blue table headers, coloured top edges on number tiles and an orange warning box for low attendance.
- **Programs (management):** each programme is a card with a coloured banner and short code (CE, EE, IT …); two cards per row on phones.
- **Teachers table:** the “Class” column always showed “A” (the section name), which meant nothing. It now shows the number of students in each module, and the hours column also shows how many classes were held.

# 3. System flow

This section explains how CheckIn works from start to finish, so the whole team describes it the same way in the sprint and the demo.

## 3.1 Attendance flow

Attendance is timetable based: every counted class is either a timetable class or an extra class booked by the tutor. The diagram shows who does what; the numbered steps are explained below it.

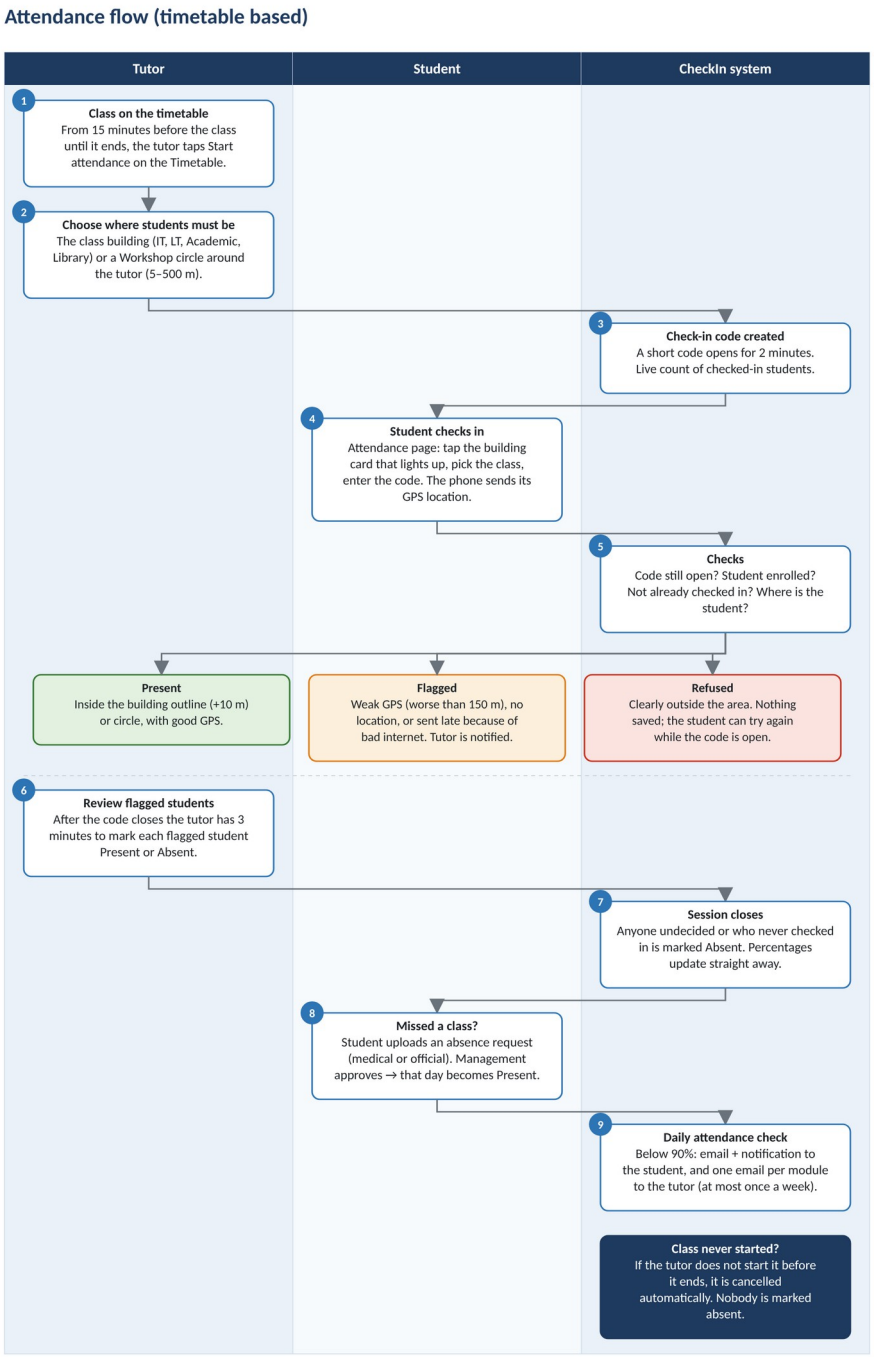



*Figure 1 — Attendance flow*

1. **Start:** from 15 minutes before a class until it ends, the tutor taps Start attendance on the Timetable (or Start Attendance page). At any other time the tutor books an extra class first.
2. **Location:** the tutor chooses the class building, or a Workshop circle around their own position for labs and unmapped rooms.
3. **Code:** a short check-in code opens for 2 minutes and the tutor sees a live count.
4. **Check in:** the student taps the building card that lights up on the Attendance page, picks the class and types the code; the phone waits a few seconds for a good GPS reading.
5. **Checks:** the server confirms the code is open, the student is enrolled and not already checked in, then compares the location with the geofence: Present, Flagged or Refused.
6. **Review:** flagged check-ins do not count until the tutor accepts them. After the code closes the tutor has 3 minutes to decide.
7. **Close:** undecided and missing students become Absent; the student's % (hours attended ÷ hours held) updates straight away.
8. **Evidence:** a student who missed a class can upload a medical or official absence request; when management approves it, that day's classes become Present.
9. **Alerts:** once a day students under 90% get an email and a notification, and each tutor gets one email per module.

| Rule | Value |
|---|---|
| Start window | 15 minutes before the class until it ends |
| Check-in code open | 2 minutes |
| Tutor review of flagged students | 3 minutes after the code closes |
| Building geofence | Building outline from the map + 10 m |
| Workshop geofence | Circle around the tutor, 5–500 m (default 30 m) |
| GPS accuracy tolerance | 150 m (worse readings are flagged) |
| Required attendance / at risk | 90% / below 80% |
| Class not started by its end | Cancelled automatically; nobody marked absent |
| Optional classes | Attendance taken but never counted |

## 3.2 Tutor – admin interaction

The admin (ICT Officer) controls who can teach and what they teach each semester. The tutor runs their own classes. Management watches attendance, approves absence requests and sees every class change.

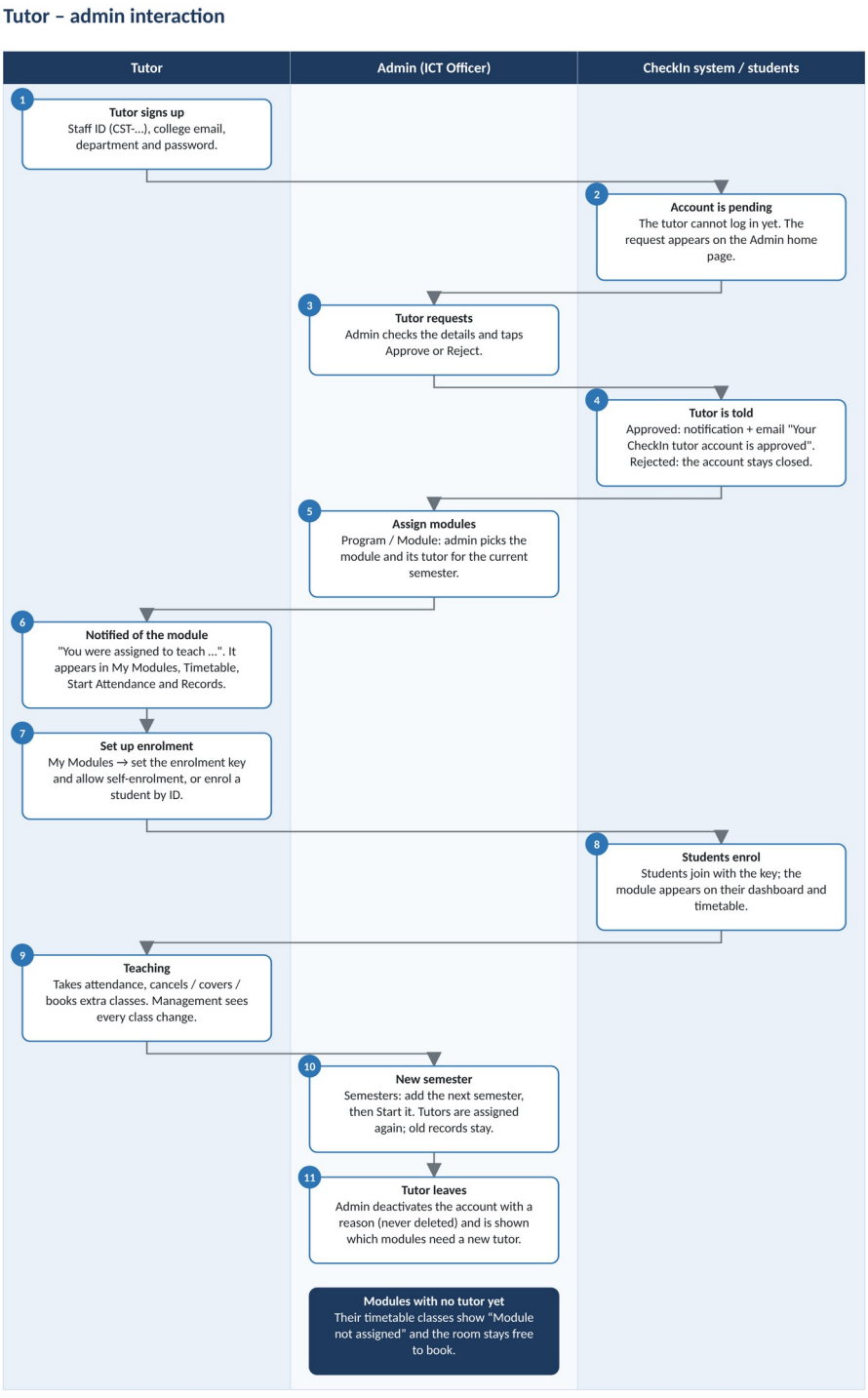



*Figure 2 — Tutor – admin interaction*

| Step | Tutor | Admin |
|---|---|---|
| Joining | Signs up with staff ID, college email and department | Approves or rejects the request on the Admin home page |

| Step | Tutor | Admin |
|---|---|---|
| Each semester | Gets a notification for each module; sets the enrolment key | Assigns a tutor to each module for the current semester |
| Teaching | Takes attendance; cancels, covers or books extra classes | — (management sees class changes) |
| New semester | Is assigned again; old records stay viewable | Adds the next semester, then starts it |
| Leaving | — | Deactivates the account with a reason (never deleted) and reassigns the modules |

### 3.3 Notifications

Every notification appears in the app (bell on the student dashboard). The ones below are also emailed to the sign-in email when the user's email switch is on. SMS is not used because the college has no SMS gateway for the project.

| Event | In-app | Email |
|---|---|---|
| Tutor account approved | Tutor | Tutor |
| Assigned to a module | Tutor | — |
| Flagged check-in needs a decision | Tutor | — |
| Flagged check-in decided | Student | — |
| Class cancelled, covered, extra or called off | Students (and cover tutor) | — |
| Absence request sent | Tutors | — |
| Absence request approved / rejected | Student and tutors | Student and tutors |
| Attendance below 90% | Student and tutor | Student and tutor |
| Password changed / reset | Account owner | Account owner |

## 4. How the work was tested

- Ran the full app locally against a copy of our database and took screenshots of every page for all four roles, on a laptop screen and a phone screen.
- Emails were tested with a local mail catcher: password change, evidence approval/rejection and the low-attendance email all arrived; with the switch off nothing was sent.
- Automatic cancellation: classes that ended without a session were cancelled once; classes with attendance, covers or extra classes were left alone; running it again created no duplicates.
- Unassigned modules: the free-period check showed a room used only by an unassigned module as free to book.
- Semesters: adding a semester that overlaps Autumn 2026 is refused with a clear message.
- Pulled the work onto my laptop as one bundle and confirmed a clean fast-forward on feature/vle-theme.

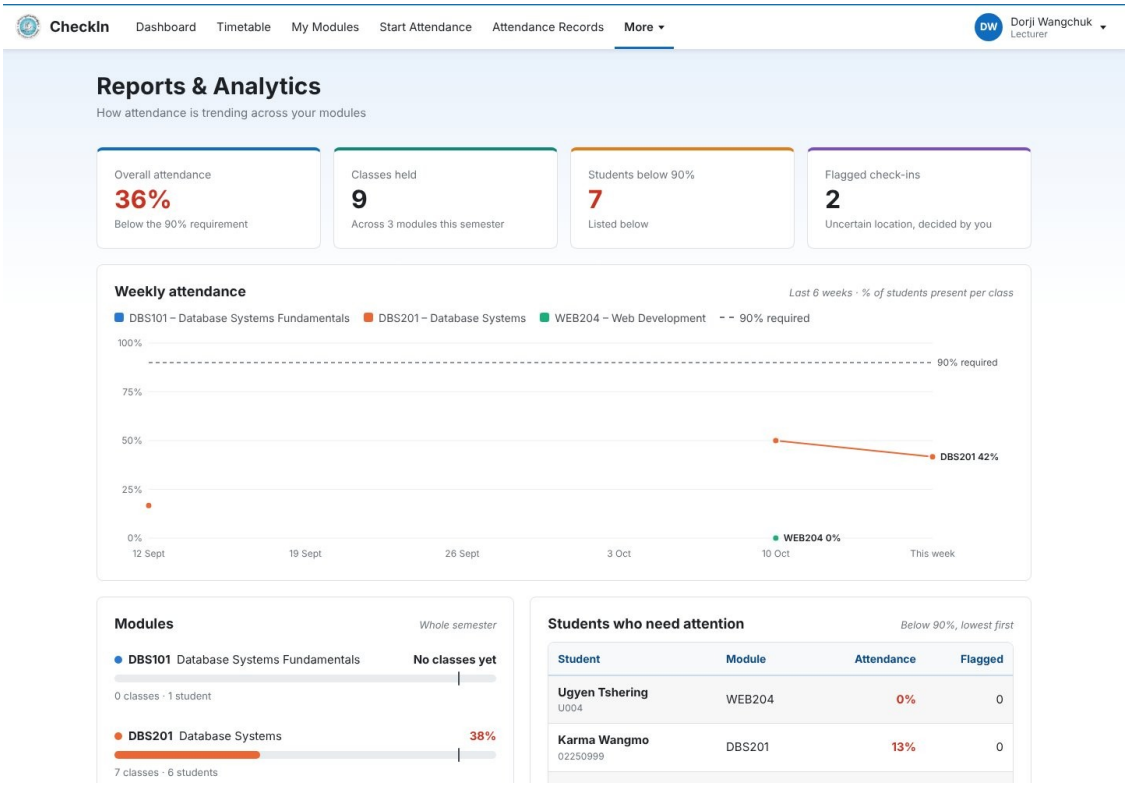
## 5. Good practices followed

- No database changes were needed, so nothing was run on Supabase.
- Login, sign-up and the landing page were not touched, as agreed.
- No secrets were committed; email settings stay in .env.
- All work is on a feature branch and stays local until the team approves it; it will go into dev through a pull request.

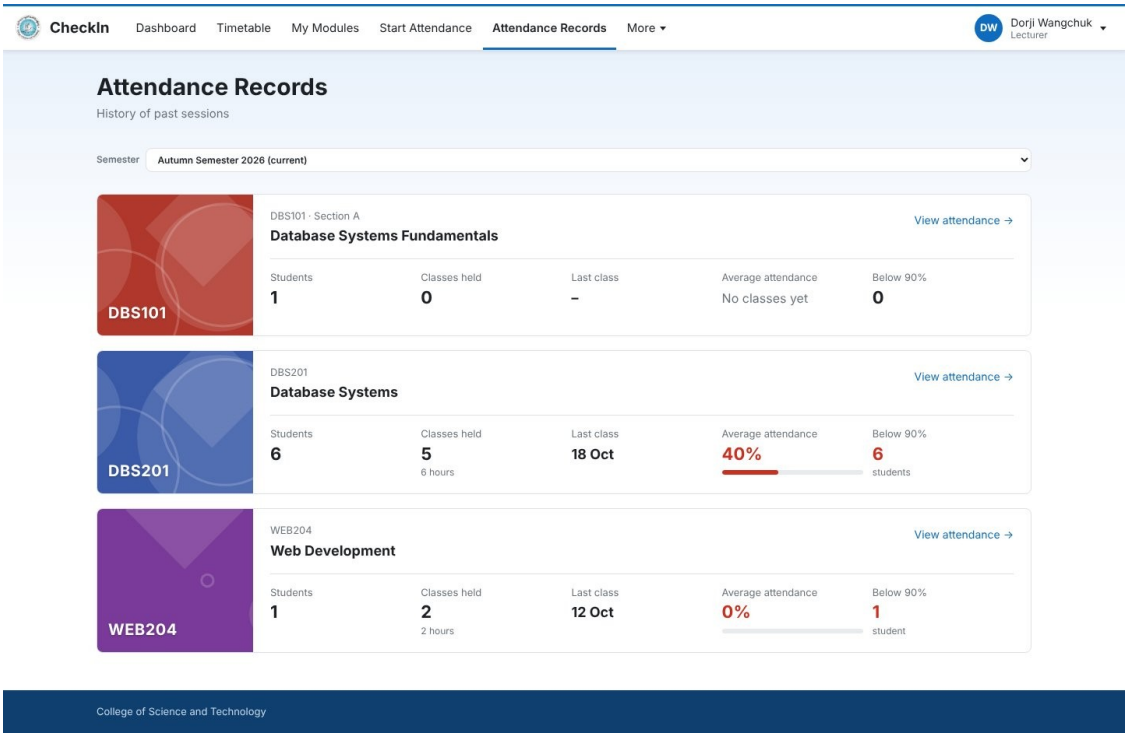
## 6. Next steps

- Team reviews the new design locally and approves it → push feature/vle-theme and open a pull request into dev.
- Correct the Spring 2027 semester dates in Supabase (the old row overlaps Autumn 2026).
- Set real Gmail SMTP settings in each backend .env so emails reach inboxes.
- Team test on phones: start attendance, check in, flagged review, and an automatically cancelled class.
- Update the README with automatic cancellation and “Module not assigned”.

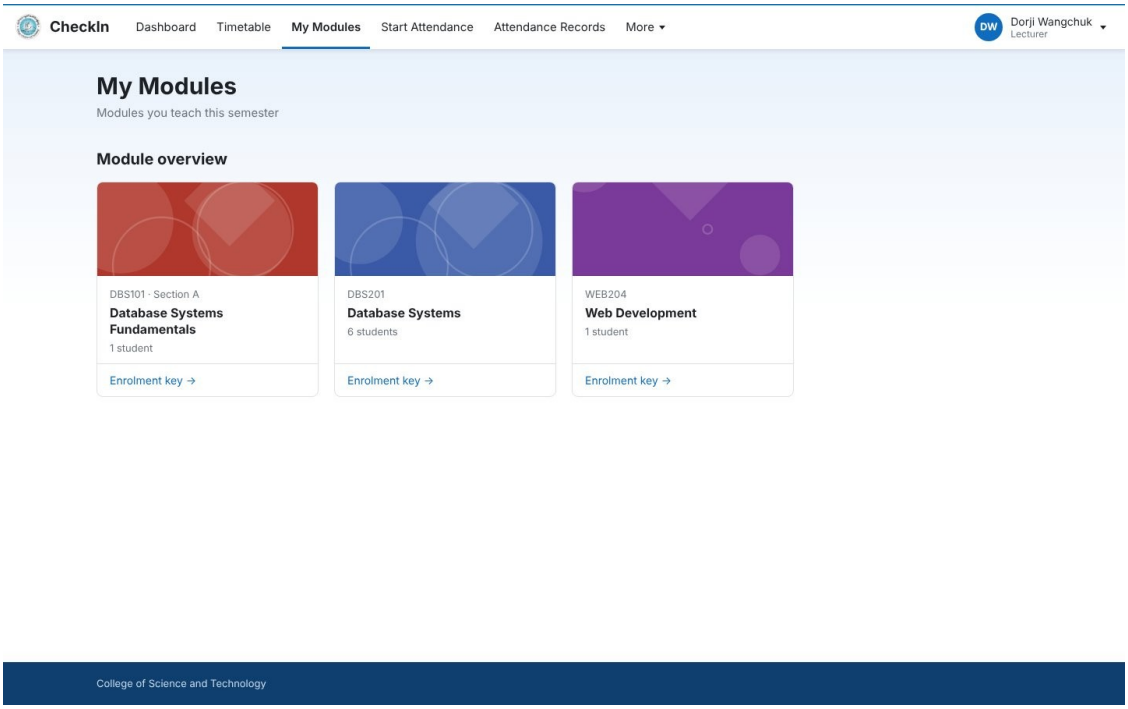
# Appendix A: Screenshots



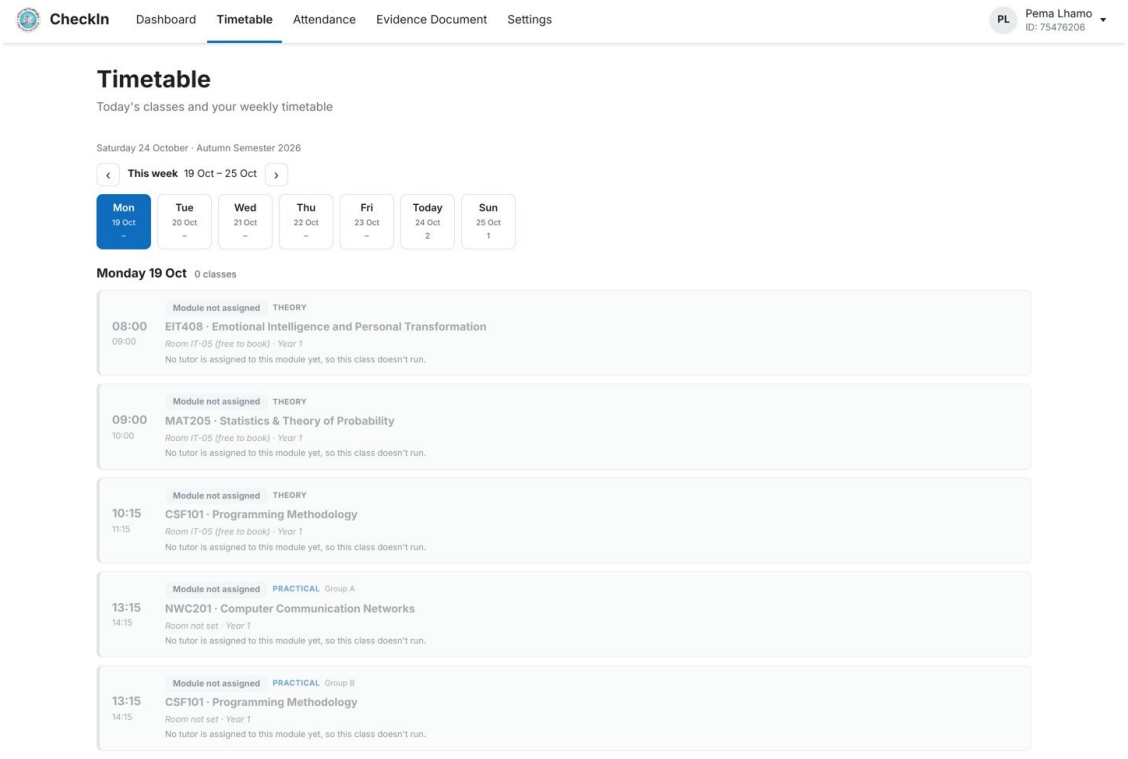

*Tutor — Reports & Analytics*



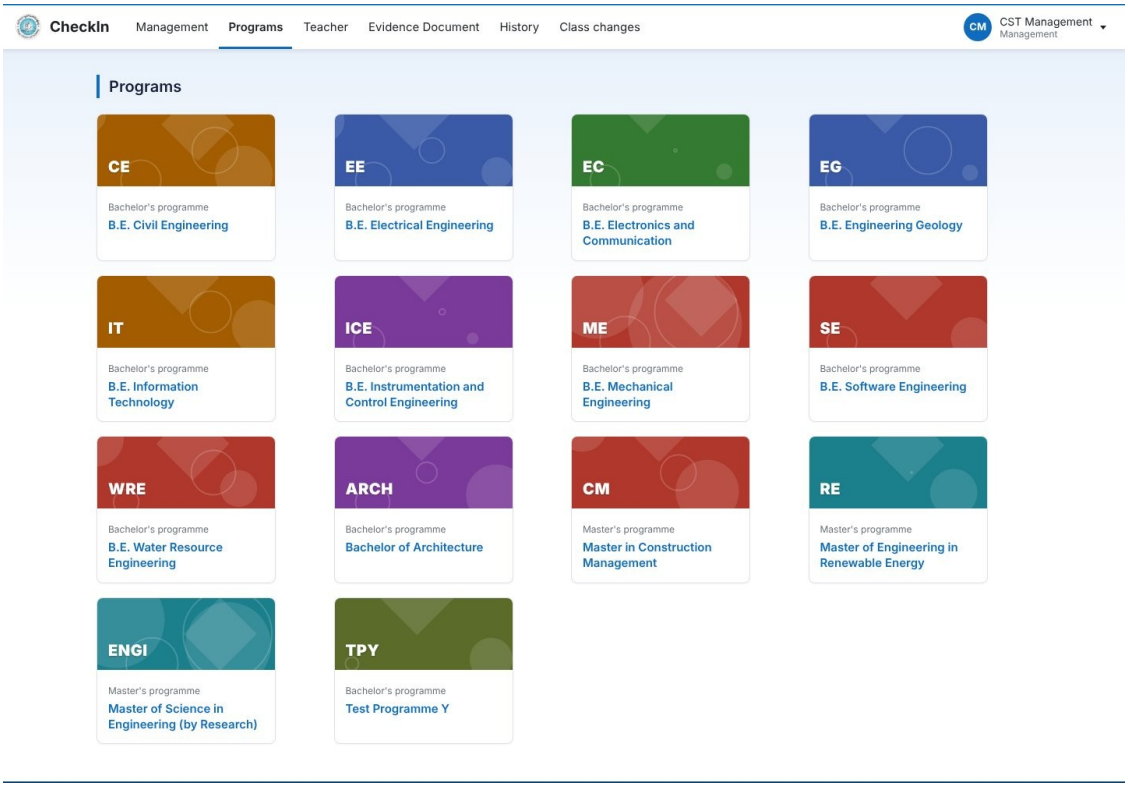

*Tutor — Attendance Records*



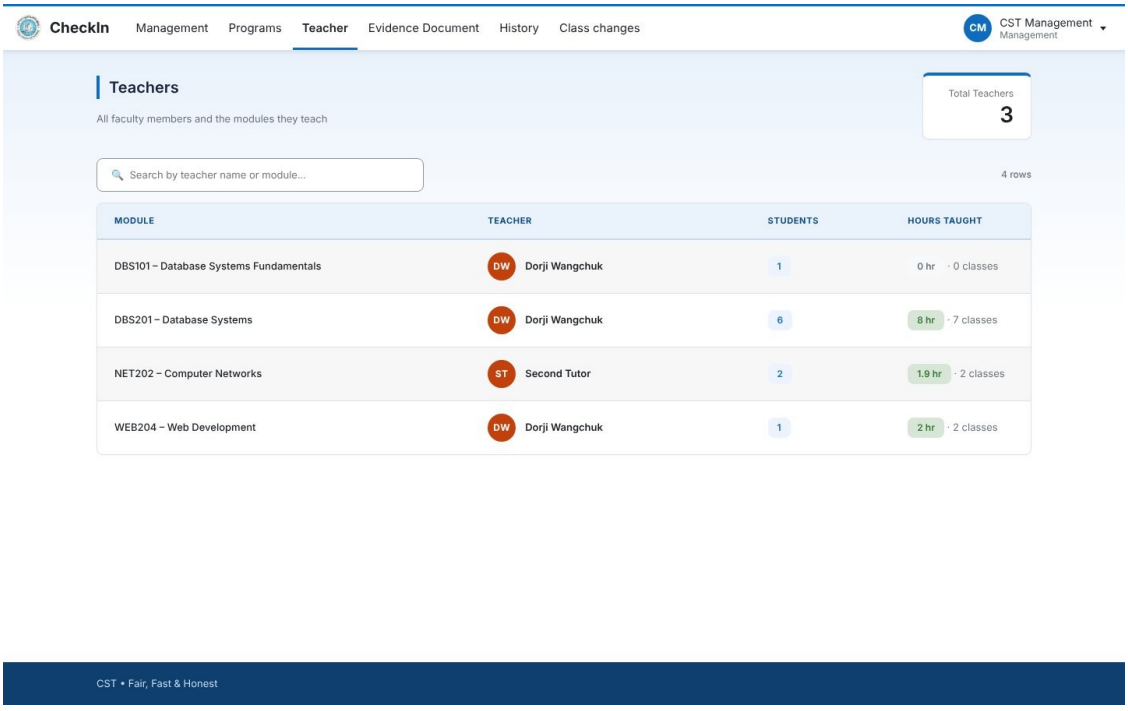

*Tutor — My Modules*




*Student — Timetable with "Module not assigned" classes*




*Management — Programs*




*Management — Teachers*

# Appendix B: My commits today

| Time | Commit | Description |
|---|---|---|
| 23:23 | b156e44 | VLE look for the logged-in app (all four roles) |
| 23:34 | 473771d | Tutor email notifications actually work; remove SMS/push toggles that did nothing |
| 23:39 | 85d1613 | My Modules in the VLE style: course-overview cards and a Moodle-style enrolment form |
| 23:45 | a13c0b6 | Attendance Records: one wide VLE-style card per module with its numbers |
| 23:51 | c9d2061 | Reports & Analytics: headline tiles, weekly trend with 90% target, module bars, students below 90% |
| 23:54 | d471124 | Absence request detail: VLE layout with status, details, reason and document preview |
| 00:01 | a6631d9 | Semesters: only ended semesters show as Past; reject overlapping semester dates |
| 00:01 | 2706864 | User menu: no Profile link for admin and management |
| 00:16 | 2b2480a | Timetable-based classes: auto-cancel classes nobody started; unassigned modules don’t block rooms |
| 00:27 | 1653073 | Colour accents across dashboards; programme cards; Teachers table shows students |